Ashton Gomes

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Education

Stony Brook University - MS in Mechanical Engineering | GPA: 3.67/4.0 — Expected May 2027

Stony Brook University - BE in Mechanical Engineering | GPA: 3.57/4.0 — Graduated May 2026

Skills

CAD & Analysis: SolidWorks, Fusion 360, Siemens NX, Finite Element Analysis (FEA), Tolerancing

Programming & Robotics: Python, C++, MATLAB, Simulink, MuJoCo, ROS/ROS2, Git, Linux

Engineering Methods: Suspension Kinematics, Robot Kinematics, Dynamics, Motion Planning, PID Control, DFM, DFA

Manufacturing & Instrumentation: CNC Machining, Milling, Turning, Soldering, Data Acquisition, LabVIEW, NI-DAQ, UART

Work and Research Experience

Graduate Researcher - Interacting Robotic Systems Laboratory - Stony Brook, NY — Jan 2026 - Present

- Develop a C++ task-space planner for a 7-DOF Kinova Gen3 arm using goal-directed RRT and dual-quaternion screw linear interpolation (ScLERP), achieving 108 ms median planning time across the MuJoCo trials passing collision and joint-limit checks.
- Implement resolved-rate motion control using dual-quaternion forward kinematics, a damped Jacobian pseudoinverse, and quaternion sign alignment to track end-effector pose targets, achieving final position error below 0.004 mm across the 23 validated simulation trials.
- Benchmark the pipeline across 40 reachable-goal trials in empty and randomized obstacle scenes, achieving 97.5% pose-target completion and evaluating generated trajectories with 240 collision and joint-limit checks per path segment.

Vehicle Dynamics Lead - Stony Brook Motorsports - Stony Brook, NY — June 2025 - Present

- Lead a 12-member team responsible for steering, suspension, and braking systems, coordinating design reviews, FEA, and fabrication schedules to complete all three systems by the end of the academic year.
- Reduced weight by approximately 0.8 lb per control arm and FEA-predicted deformation by 40% by relocating shock mounts to the upper control arms and reducing tube wall thickness from 0.085 to 0.049 in. Eliminated tube bends and halved weld-related heat-affected regions to simplify fabrication and improve weld access.
- Reduced inside turning radius from 9 ft to 2 ft by replacing 40% Ackermann geometry with 20% anti-Ackermann and incorporating passive rear steer, contributing to the team's 9th-place finish in the overall dynamic-event standings.
- Developed a MATLAB suspension-kinematics and damping models to guide motion-ratio and damping selection, alongside revisions to roll-center geometry, roll stiffness, and anti-squat that improved cornering and acceleration traction, helping the team advance from 40th to 18th in acceleration and place 18th in hill climb.

Project Management Intern - Dormitory Authority of the State of New York - New York, NY — May 2025 - Aug 2025

- Tracked a $4.4M renovation for the Dormitory Authority of the State of New York, comparing field installations with engineering drawings and specifications to identify discrepancies and monitor milestones.
- Conducted MEP field inspections to assess building-code compliance and document safety risks.

Technical Projects

Senior Design - Shock Dynamometer — Aug 2025 - May 2026

- Designed and fabricated a crank-driven shock dynamometer with a 3 HP motor, 20:1 worm gearbox, and variable-frequency drive, providing a 4-inch stroke and targeting 15 in/s peak velocity for damping characterization.
- Evaluated the frame and whiffletree load-isolation assembly under a 750 lbf worst-case load using SolidWorks FEA, obtaining factors of safety of 3.1 and 5.8, respectively.
- Developed Teensy 4.1 firmware and a Python analysis pipeline integrating a 24-bit ADS1220 ADC and quadrature encoder to capture force and calculate axial velocity, generating compression/rebound force-velocity curves and estimating branch slopes through median binning and cross-validated fitting.

Crazyflie Autonomous Navigation and Landing — Sep 2026 - Present

- Implemented range-based obstacle detection and randomized avoidance maneuvers, completing 8 of 10 navigation trials without collisions.
- Developed boundary recovery logic using optical-flow surface quality to detect positioning uncertainty, returning the drone to the operating area in 9 of 10 boundary encounters.
- Localized landing pads by detecting height changes and mapping pad edges, achieving a mean landing error of 7 cm from the pad center across 10 trials.